IF study – Final Presentation

March 30, 2023

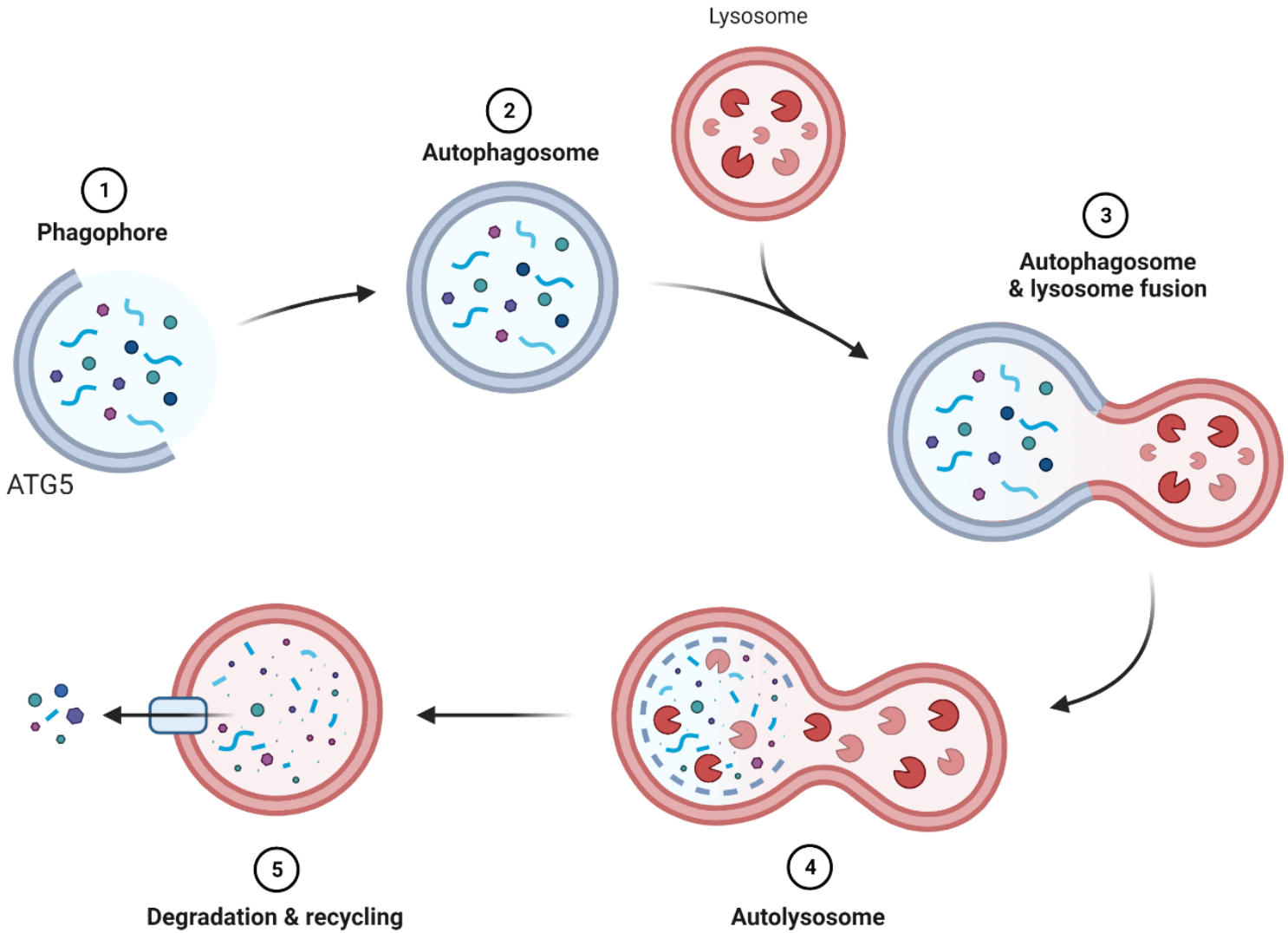
Nathalie Costie

Feasibility Study of Intermittent Fasting in Chronic Lymphocytic Leukemia and Small Lymphocytic Lymphoma (CLL/SLL) Patients at BC Cancer - Victoria



Study team:
Research Lead – Eleah Stringer, MSc, RD, CSO
PI – Dr. Nicol Macpherson, Medical Oncologist
Co-Investigator – Dr. Julian Lum
Research Assistants: Nathalie Costie & Samantha Punch

BC CANCER FOUNDATION


Lysosome
2
Autophagosome
1
Phagophore
3
Autophagosome
& lysosome fusion
ATG5
5
Degradation & recycling
4
Autolysosome

Dual role of autophagy in cancer

- In **premalignant lesions**, animal models suggest that autophagy **might prevent cancer development** by protecting cells from oxidative stress, DNA damage, inflammation and the accumulation of dysfunctional organelles (Antunes et al., 2018; Galluzzi et al., 2015).

- **In contrast**, once cancer is established, autophagy **may promote tumor cell survival and growth** by allowing tumors to cope with stressors including hypoxia, nutritional and growth factor deprivation, and chemotherapy allowing the tumors to adapt, proliferate, survive and disseminate (Lum et al., 2005); Levine, Mizushima, & Virgin, 2011; Amaravadi et al., 201; White, 2012)

Intermittent fasting, autophagy, and cancer

- Limited findings however…
  - *Augment intracellular protein quality (Lee et al., 2014)*
  - *Improves the maintenance of the proteome by targeting autophagy-dependent TORC1*

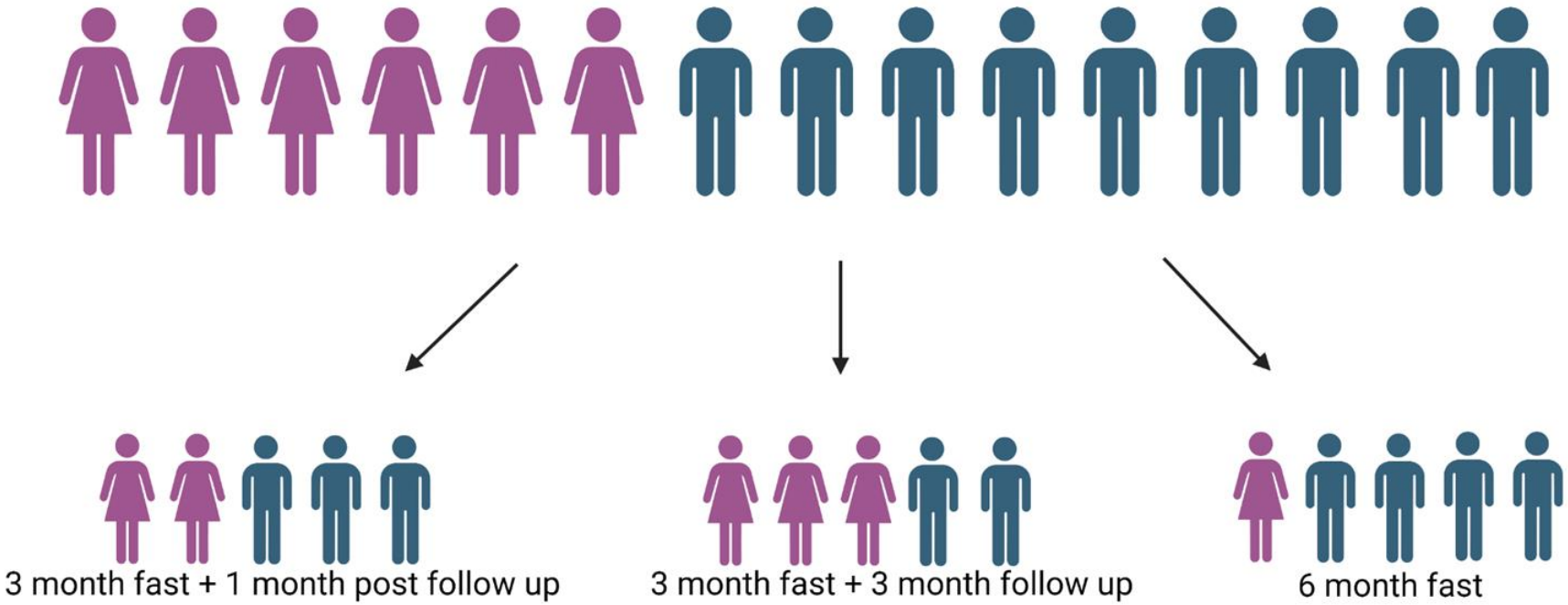
- Calorie Restriction has shown to downregulate TORC1 & upregulate autophagy in a number of tissues (Towers & Thorburn, 2016)
  - *Restore autophagic function & thereby preserve organelle quality (Stockman et al., 2018)*

Autophagy Flux Data

INCLUDED: IF-002, 004, 006, 007, 009, 010, 013, 017, 018, 019, 020, 021, 022

EXCLUDED: IF-005, IF-012 (partial time points, suboptimal protein collection)

Question: does intermittent fasting upregulate autophagy in patients with CLL/SLL?



5 patients consented to the initial 4 month study

- IF-002, 004, 005, 006, 007

10 patients consented to the 6 month extension

- 5 patients opted to fast for 6 months (IF-017, 019, 020, 021, 022)
- 5 patients ended the fast after 90 days (IF-009, 010, 012, 013, 018)

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Autophagy Flux Assay Workflow

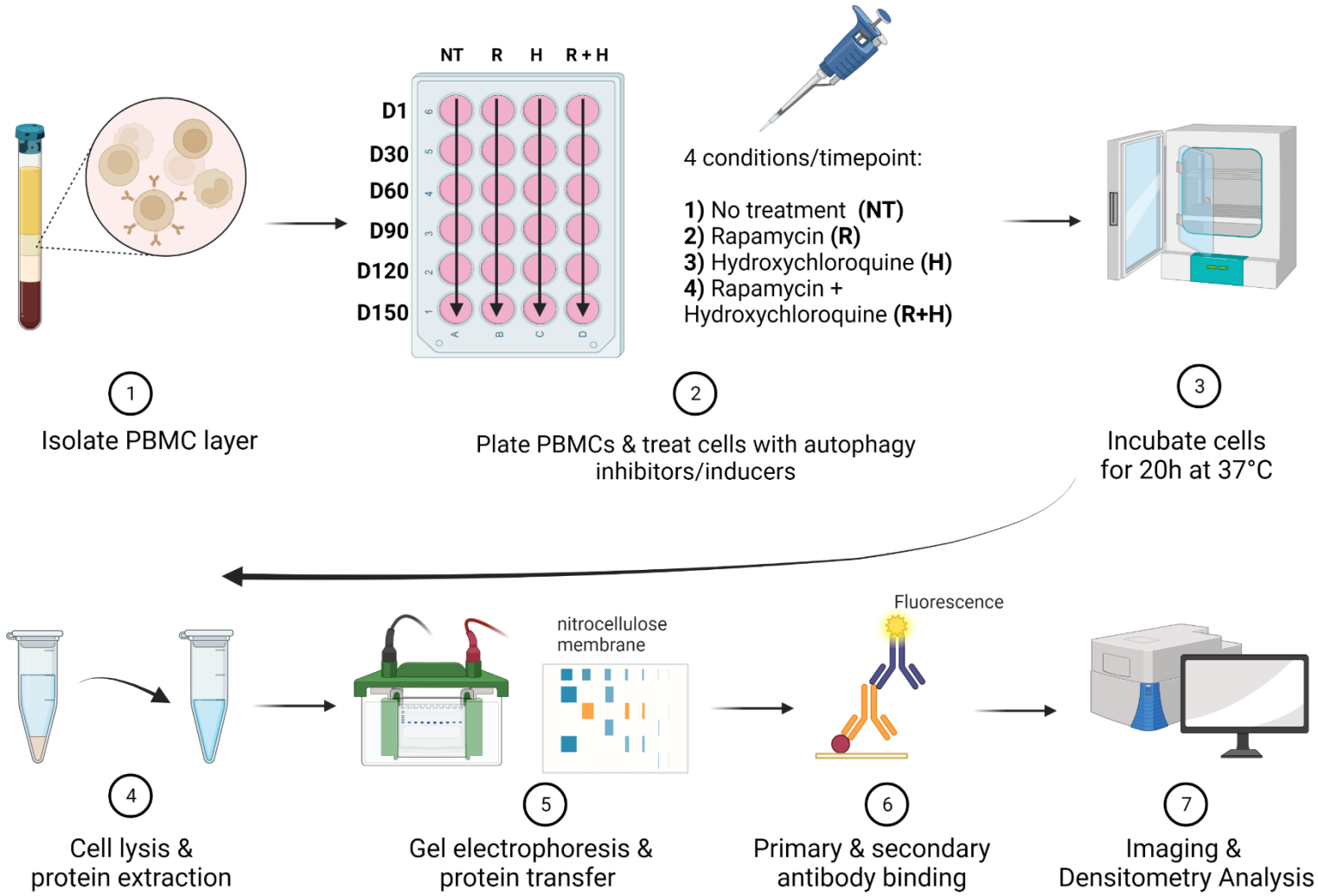
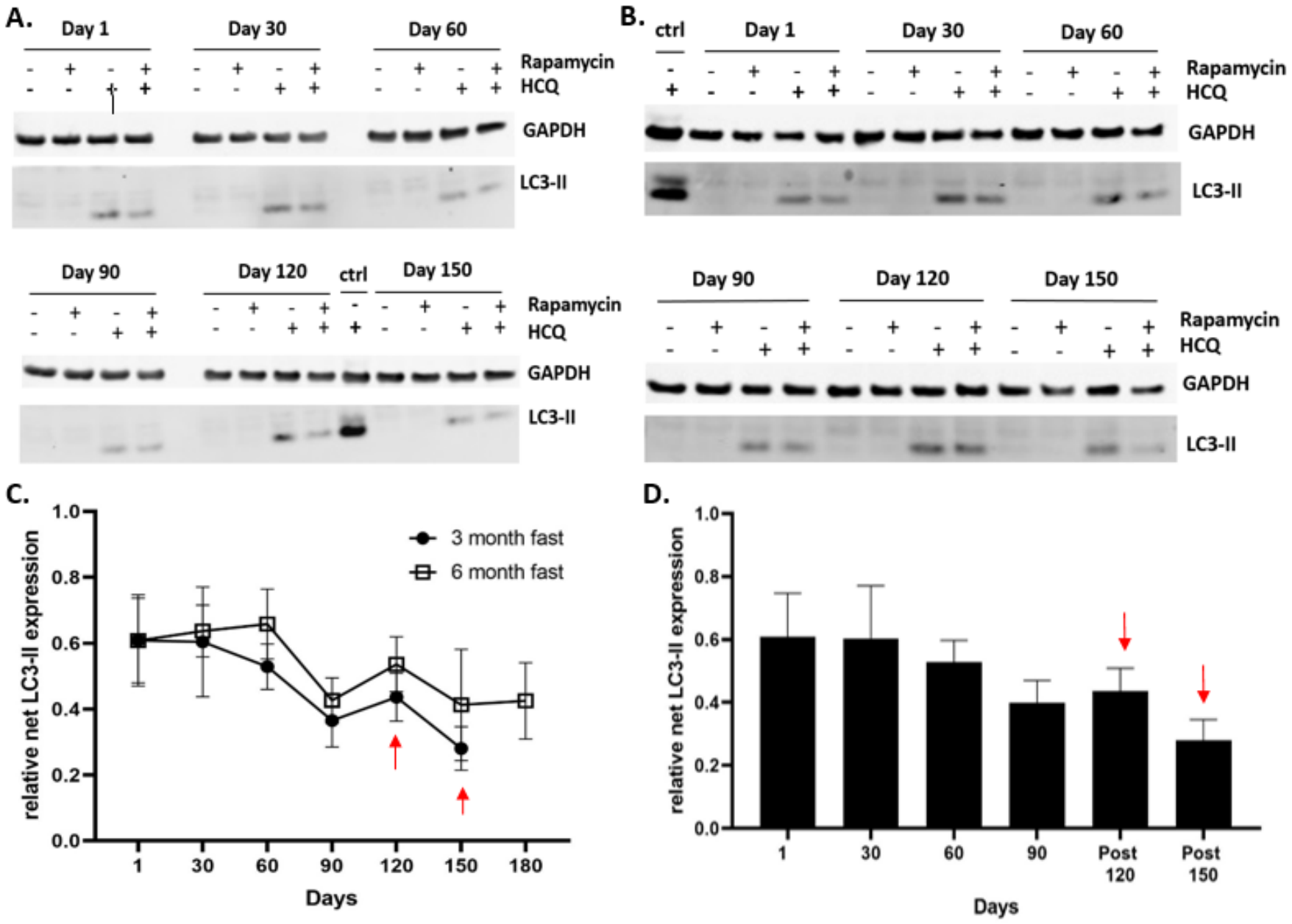


Image created with BioRender.com


A.
Day 1
Day 30
Day 60
Rapamycin
HCQ
GAPDH
LC3-II
Day 90
Day 120
ctrl
Day 150
B.
ctrl
C.
3 month fast
6 month fast
relative net LC3-II expression
Days
D.
Post 120
Post 150

Metabolomics Data

INCLUDED: IF-002, 004, 006, 007, 009, 010, 013, 017, 018, 019, 020, 021, 022

Question: will we see any fasting-induced changes in metabolites?

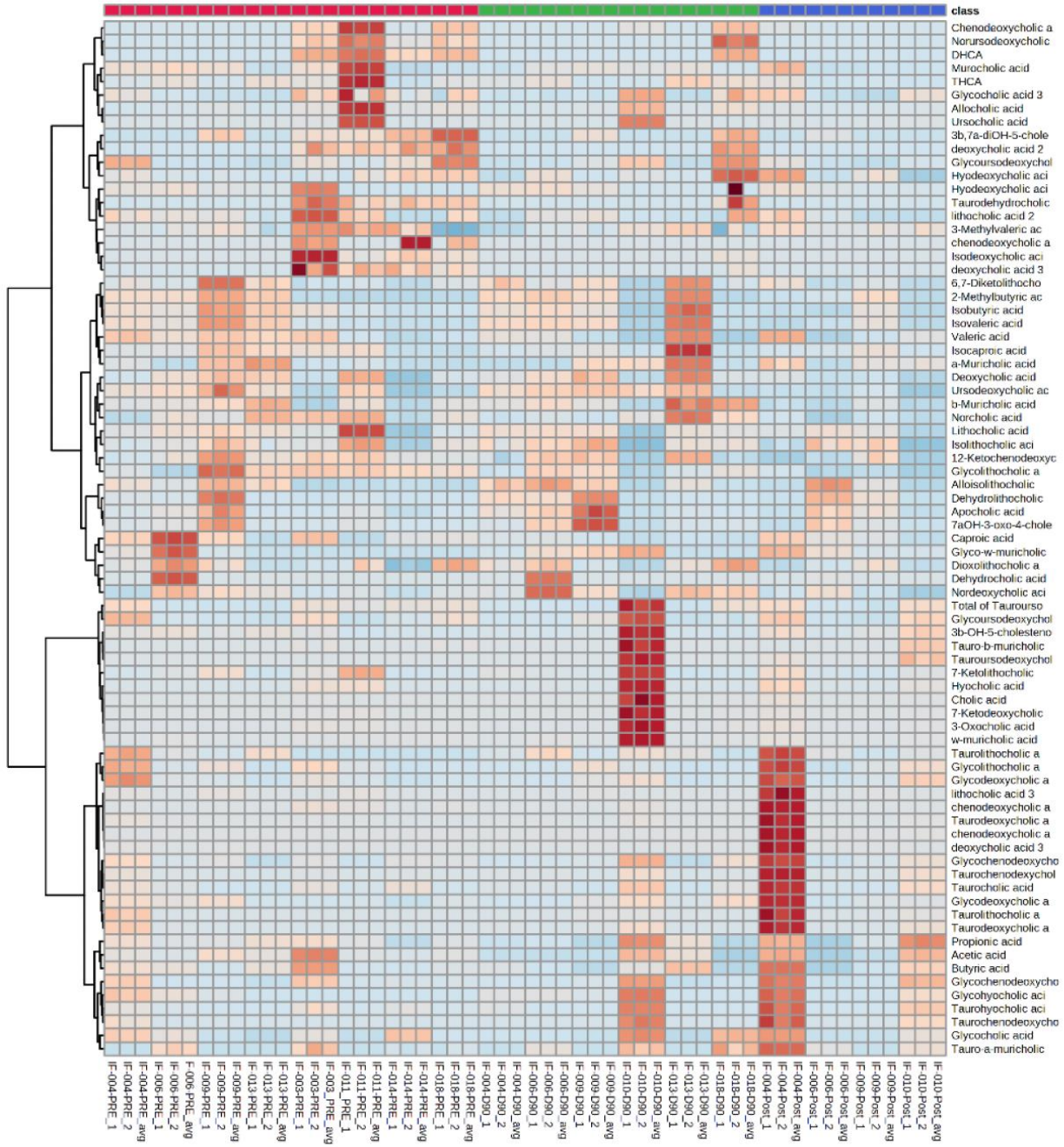
Bile acids & SCFAs

- The bile acid (BA) pool consists of **primary** BAs, synthesized by hepatocytes and stored in the gall bladder, and when **secreted into the intestine**, primary BAs are metabolized into **secondary** BAs **by intestinal bacteria**
- **Short-chain fatty acids (SCFAs)** are the main metabolites produced in the colon and are the products of fermentation of insoluble fibre (e.g. cellulose, resistant starch) from diet by the bacteria in the gut and are shown to modulate gut inflammation.

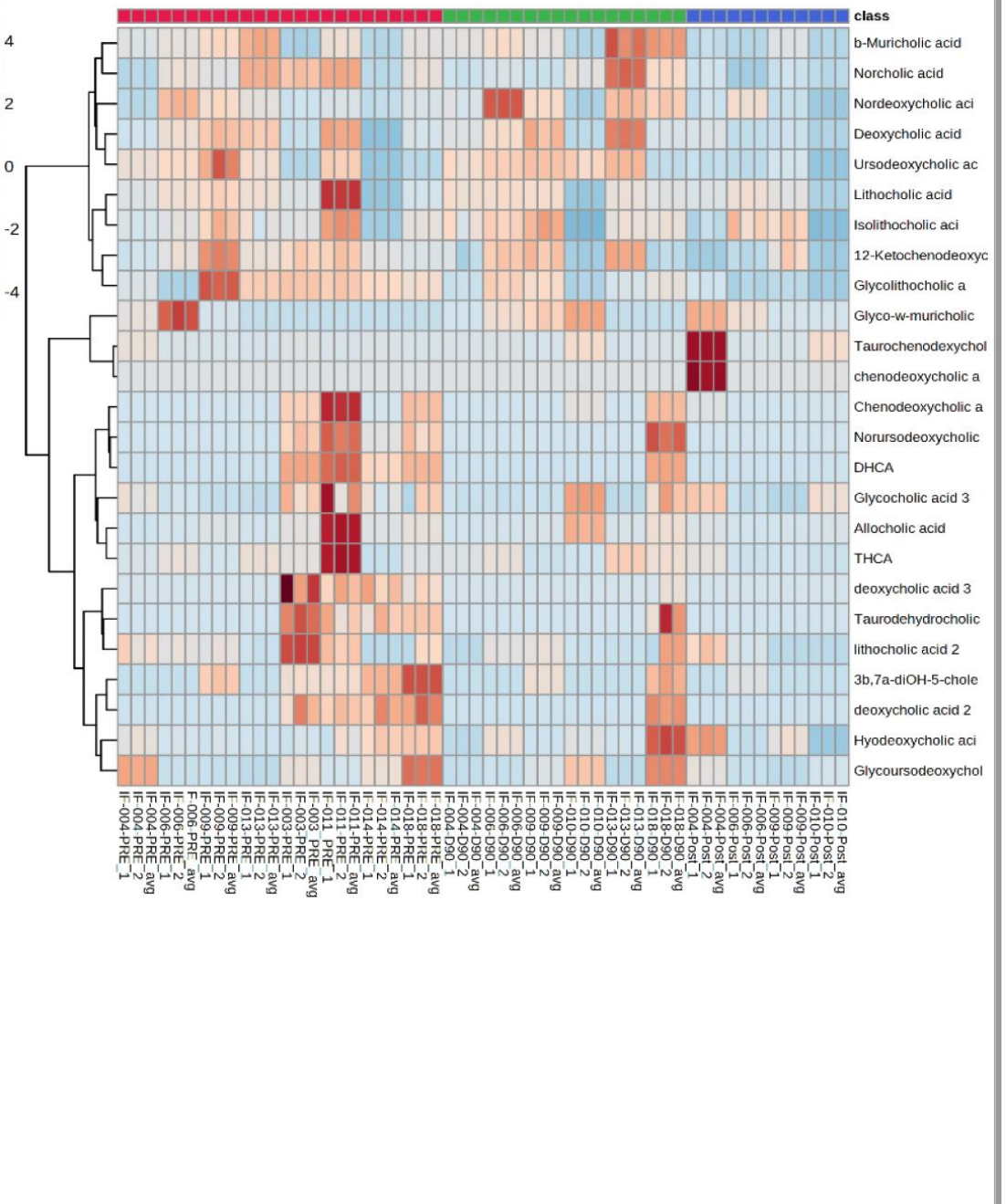
3-month intermittent fasting diet group

INCLUDED: IF-002, 003, 004, 006, 007, 009, 010, 011, 013, 014

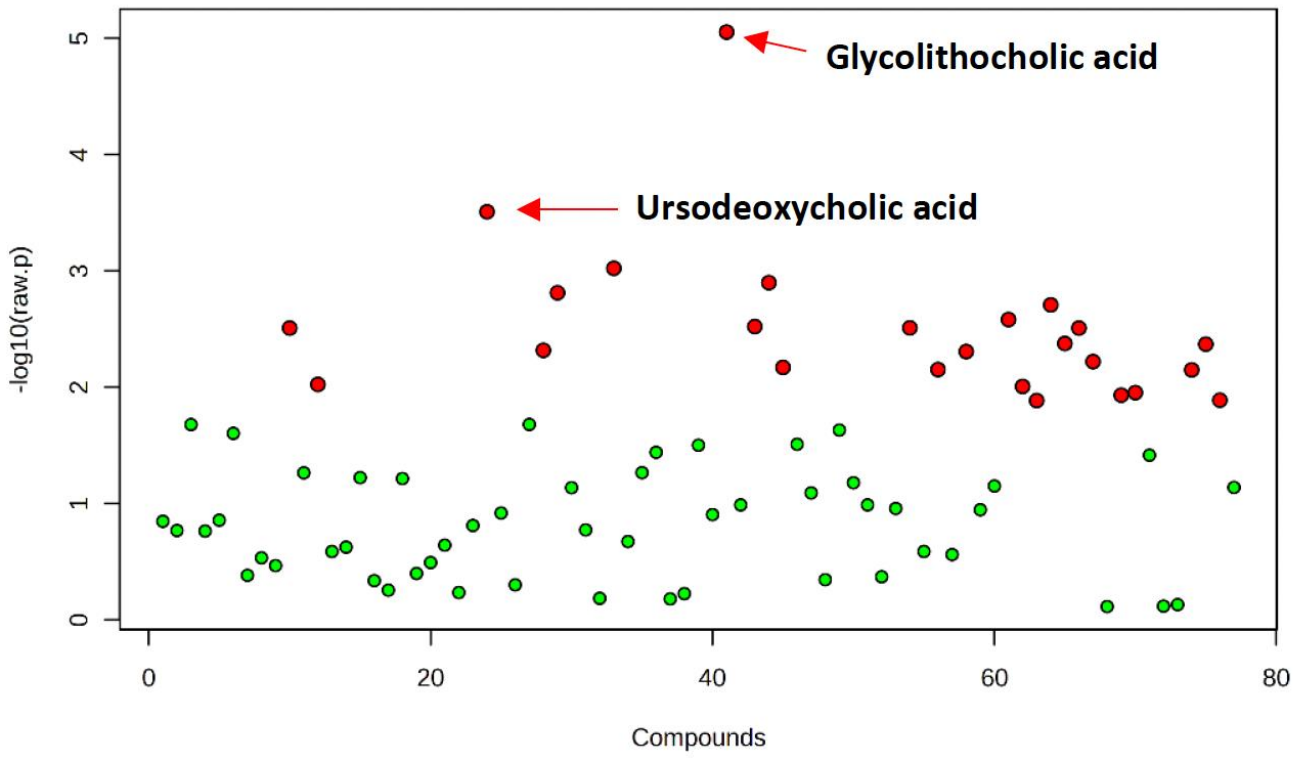
A.


class
Chenodeoxycholic a
Norursodeoxycholic
DHCA
Murocholic acid
THCA
Glycocholic acid 3
Allocholic acid
Ursocholic acid
3b,7a-diOH-5-chole
deoxycholic acid 2
Glycoursodeoxychol
Hyodeoxycholic aci
Hyodeoxycholic aci
Taurodehydrocholic
lithocholic acid 2
3-Methylvaleric ac
chenodeoxycholic a
Isodeoxycholic aci
deoxycholic acid 3
6,7-Diketolithocho
2-Methylbutyric ac
Isobutyric acid
Isovaleric acid
Valeric acid
Isocaproic acid
a-Muricholic acid
Deoxycholic acid
Ursodeoxycholic ac
b-Muricholic acid
Norcholic acid
Lithocholic acid
Isolithocholic aci
12-Ketochenodeoxyc
Glycolithocholic a
Alloisolithocholic
Dehydrolithocholic
Apocholic acid
7aOH-3-oxo-4-chole
Caproic acid
Glyco-w-muricholic
Dioxolithocholic a
Dehydrocholic acid
Nordeoxycholic aci
Total of Taurourso
Glycoursodeoxychol
3b-OH-5-cholesteno
Tauro-b-muricholic
Tauroursodeoxychol
7-Ketolithocholic
Hyocholic acid
Cholic acid
7-Ketodeoxycholic
3-Oxocholic acid
w-muricholic acid
Taurolithocholic a
Glycolithocholic a
Glycodeoxycholic a
lithocholic acid 3
chenodeoxycholic a
Taurodeoxycholic a
chenodeoxycholic a
deoxycholic acid 3
Glycochenodeoxycho
Taurochenodexychol
Taurocholic acid
Glycodeoxycholic a
Taurolithocholic a
Taurodeoxycholic a
Propionic acid
Acetic acid
Butyric acid
Glycochenodeoxycho
Glycohyocholic aci
Taurohyocholic aci
Taurochenodeoxycho
Glycocholic acid
Tauro-a-muricholic
IF-004-PRE_1
IF-004-PRE_2
IF-004-PRE_avg
IF-006-PRE_1
IF-006-PRE_2
F-006 PRE_avg
IF-009-PRE_1
IF-009-PRE_2
IF-009-PRE_avg
IF-013-PRE_1
IF-013-PRE_2
IF-013-PRE_avg
IF-003-PRE_1
IF-003-PRE_2
IF-003_PRE_avg
IF-011_PRE_1
IF-011-PRE_2
IF-011-PRE_avg
IF-014-PRE_1
IF-014-PRE_2
IF-014-PRE_avg
IF-018-PRE_1
IF-018-PRE_2
IF-018-PRE_avg
IF-004-D90_1
IF-004-D90_2
IF-004-D90_avg
IF-006-D90_1
IF-006-D90_2
IF-006-D90_avg
IF-009-D90_1
IF-009-D90_2
IF-009-D90_avg
IF-010-D90_1
IF-010-D90_2
IF-010-D90_avg
IF-013-D90_1
IF-013-D90_2
IF-013-D90_avg
IF-018-D90_1
IF-018-D90_2
IF-018-D90_avg
IF-004-Post_1
IF-004-Post_2
IF-004-Post_avg
IF-006-Post_1
IF-006-Post_2
IF-006-Post_avg
IF-009-Post_1
IF-009-Post_2
IF-009-Post_avg
IF-010-Post_1
IF-010-Post_2
IF-010-Post_avg

B.


4
2
0
-2
-4
class
b-Muricholic acid
Norcholic acid
Nordeoxycholic aci
Deoxycholic acid
Ursodeoxycholic ac
Lithocholic acid
Isolithocholic aci
12-Ketochenodeoxyc
Glycolithocholic a
Glyco-w-muricholic
Taurochenodexychol
chenodeoxycholic a
Chenodeoxycholic a
Norursodeoxycholic
DHCA
Glycocholic acid 3
Allocholic acid
THCA
deoxycholic acid 3
Taurodehydrocholic
lithocholic acid 2
3b,7a-diOH-5-chole
deoxycholic acid 2
Hyodeoxycholic aci
Glycoursodeoxychol
IF-004-PRE_1
IF-004-PRE_2
IF-004-PRE_avg
IF-006-PRE_1
IF-006-PRE_2
F-006 PRE_avg
IF-009-PRE_1
IF-009-PRE_2
IF-009-PRE_avg
IF-013-PRE_1
IF-013-PRE_2
IF-013-PRE_avg
IF-003-PRE_1
IF-003-PRE_2
IF-003_PRE_avg
IF-011_PRE_1
IF-011-PRE_2
IF-011-PRE_avg
IF-014-PRE_1
IF-014-PRE_2
IF-014-PRE_avg
IF-018-PRE_1
IF-018-PRE_2
IF-018-PRE_avg
IF-004-D90_1
IF-004-D90_2
IF-004-D90_avg
IF-006-D90_1
IF-006-D90_2
IF-006-D90_avg
IF-009-D90_1
IF-009-D90_2
IF-009-D90_avg
IF-010-D90_1
IF-010-D90_2
IF-010-D90_avg
IF-013-D90_1
IF-013-D90_2
IF-013-D90_avg
IF-018-D90_1
IF-018-D90_2
IF-018-D90_avg
IF-004-Post_1
IF-004-Post_2
IF-004-Post_avg
IF-006-Post_1
IF-006-Post_2
IF-006-Post_avg
IF-009-Post_1
IF-009-Post_2
IF-009-Post_avg
IF-010-Post_1
IF-010-Post_2
IF-010-Post_avg

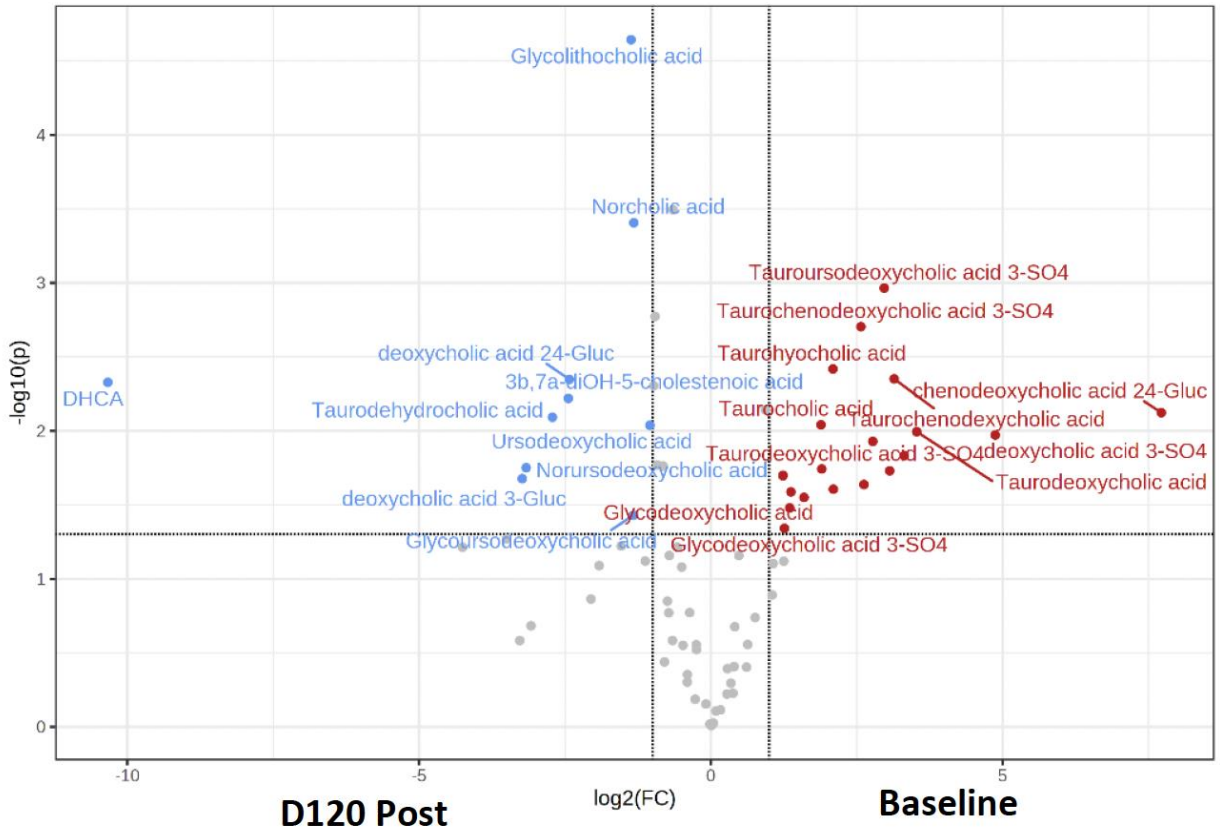
A.

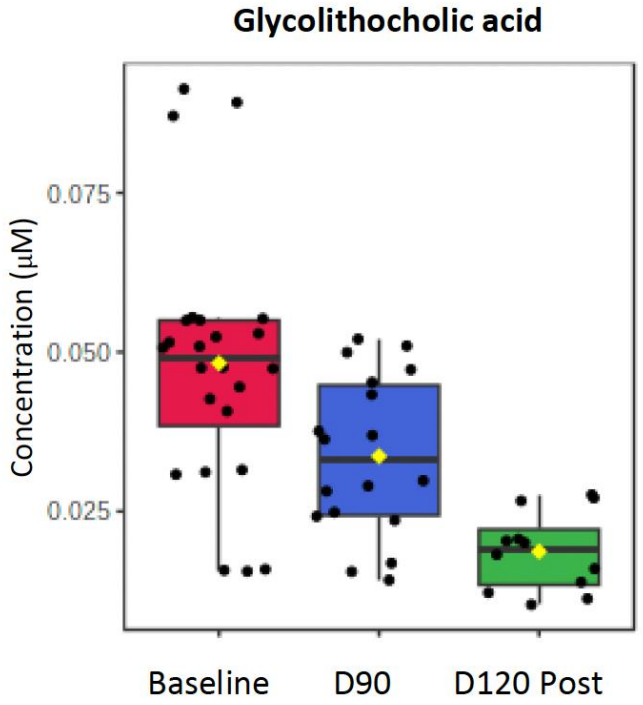


25 metabolites p<0.05*

Metabolite	f.value	p.value	-log10(p)	FDR
Glycolithocholic acid	14.747	8.83E-06	5.054	0.0006799
Ursodeoxycholic acid	9.5078	0.0003094	3.5094	0.011913
b-Muricholic acid	8.0046	0.0009476	3.0234	0.021649
Taurochenodexycholic acid	7.6327	0.0012596	2.8998	0.021649
Nordeoxycholic acid	7.3728	0.0015398	2.8125	0.021649
chenodeoxycholic acid 24-Gluc	7.0663	0.0019551	2.7088	0.021649
deoxycholic acid 3-Gluc	6.6978	0.0026134	2.5828	0.021649
Taurodeoxycholic acid	6.5232	0.003002	2.5226	0.021649
Norcholic acid	6.4911	0.0030797	2.5115	0.021649
deoxycholic acid 3-SO4	6.4859	0.0030925	2.5097	0.021649
Caproic acid	6.4859	0.0030927	2.5097	0.021649
lithocholic acid 3-SO4	6.1057	0.0041953	2.3772	0.025215
Taurodeoxycholic acid 3-SO4	6.0876	0.0042571	2.3709	0.025215
12-Ketochenodeoxycholic acid	5.9381	0.0048046	2.3183	0.025289
DHCA	5.9072	0.0049265	2.3075	0.025289
chenodeoxycholic acid 3-SO4	5.6638	0.0060074	2.2213	0.02869
Taurocholic acid	5.523	0.0067425	2.1712	0.02869
3b,7a-diOH-5-cholestenoic acid	5.4714	0.0070352	2.1527	0.02869
Taurolithocholic acid 3-SO4	5.4638	0.0070794	2.15	0.02869
Deoxycholic acid	5.1169	0.0094358	2.0252	0.036018
deoxycholic acid 24-Gluc	5.0686	0.0098231	2.0078	0.036018
Glycodeoxycholic acid 3-SO4	4.9215	0.011109	1.9543	0.038882
Glycolithocholic acid 3-SO4	4.8624	0.011674	1.9328	0.039081
Taurochenodeoxycholic acid 3-SO4	4.7434	0.012903	1.8893	0.040027
chenodeoxycholic acid 3-Gluc	4.7349	0.012996	1.8862	0.040027

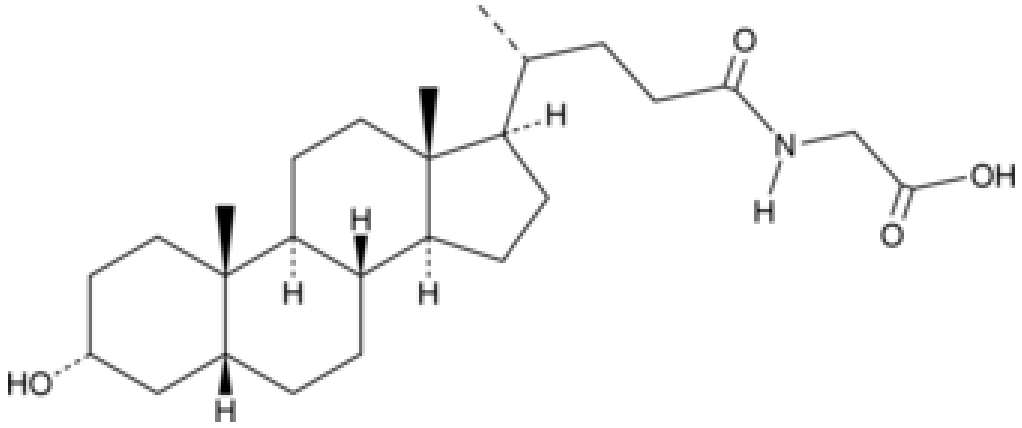
B.


Glycolithocholic acid
Norcholic acid
Tauroursodeoxycholic acid 3-SO4
Taurochenodeoxycholic acid 3-SO4
deoxycholic acid 24-Gluc
Taurohyocholic acid
3b,7a-diOH-5-cholestenoic acid
DHCA
Taurodehydrocholic acid
Taurocholic acid
Taurochenodexycholic acid
Ursodeoxycholic acid
Taurodeoxycholic acid 3-SO4
Norursodeoxycholic acid
deoxycholic acid 3-SO4
Taurodeoxycholic acid
Glycodeoxycholic acid
Glycoursodeoxycholic acid
Glycodeoxycholic acid 3-SO4
-log10(p)
log2(FC)
D120 Post
Baseline
Status
DOWN
Non-SIG
UP

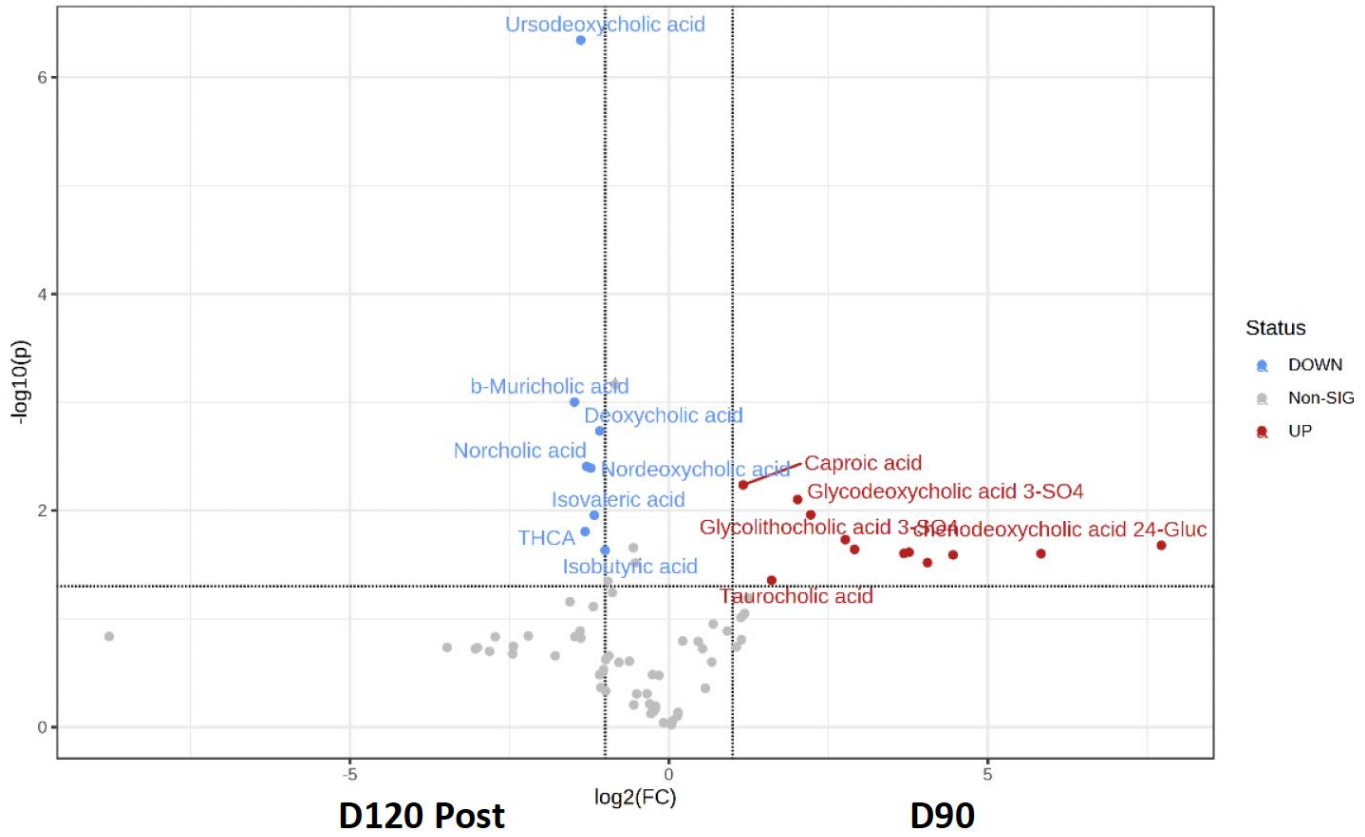

Glycolithocholic acid
Concentration (μM)
0.075
0.050
0.025
Baseline
D90
D120 Post

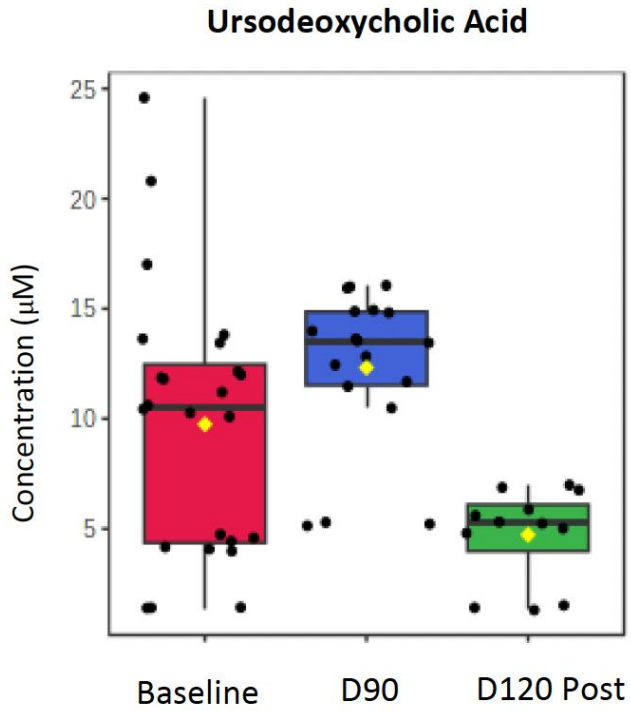
Glycolithocholic Acid

- A glycine-conjugated form of lithocholic acid
- It is increased (0.0→0.6 nmol/g) in the liver of mice fed a diet supplemented with 3% ursodeoxycholic acid (Klassan & Zhang, 2010) – *Journal of Lipid Research*
- Glycolithocholic acid levels are decreased in the plasma following subcutaneous administration of PEG-obestatin(Cys10, Cys13), a modified peptide hormone, in lean or diet-induced obese mice (Graham et al, 2016) – *Metabolomics*
- Serum glycolithocholic acid levels increase with age in children *- J Pediatr Gastroenterol Nutr.*



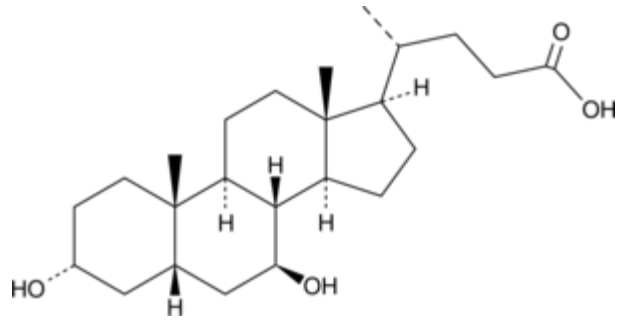
C.


Ursodeoxycholic acid
b-Muricholic acid
Deoxycholic acid
Norcholic acid
Nordeoxycholic acid
Isovaleric acid
THCA
Isobutyric acid
Caproic acid
Glycodeoxycholic acid 3-SO4
Glycolithocholic acid 3-SO4
Glycodeoxycholic acid 24-Gluc
Taurocholic acid
-log10(p)
log2(FC)
D120 Post
D90
Status
DOWN
Non-SIG
UP


Ursodeoxycholic Acid
Concentration (µM)
Baseline
D90
D120 Post

Ursodeoxycholic Acid

- Ursodeoxycholic acid (UDCA) is a secondary bile acid formed from primary BA chenodeoxycholic acid



Clinical Relevance

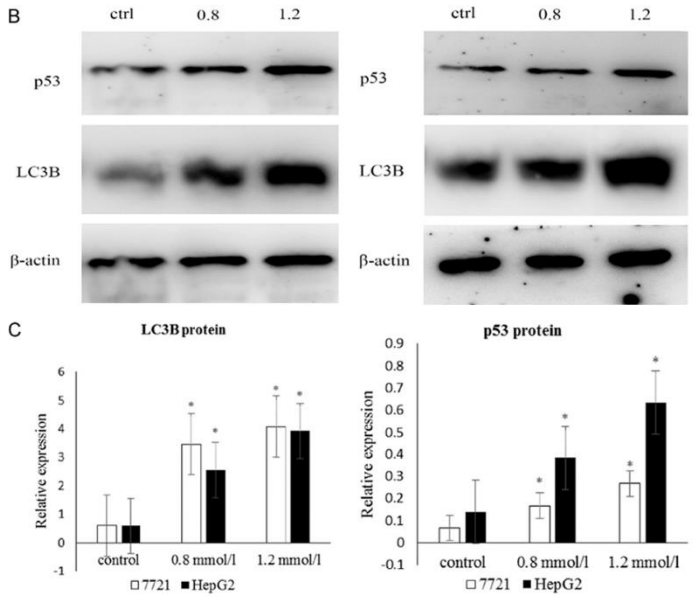
- Used to treat liver and gallstone diseases
- In a 2016 study by Patel & Seetharam, the probability of remaining free of extensive fibrosis or cirrhosis after UDCA treatment was 76% after being treated for 4 years

Mechanism

- Decreases cholesterol saturation by inhibiting the absorption of cholesterol in the intestine and its secretion into bile
- Cytoprotective effect of UDCA is attributable to its ability to protect hepatocytes from bile acid-induced damage

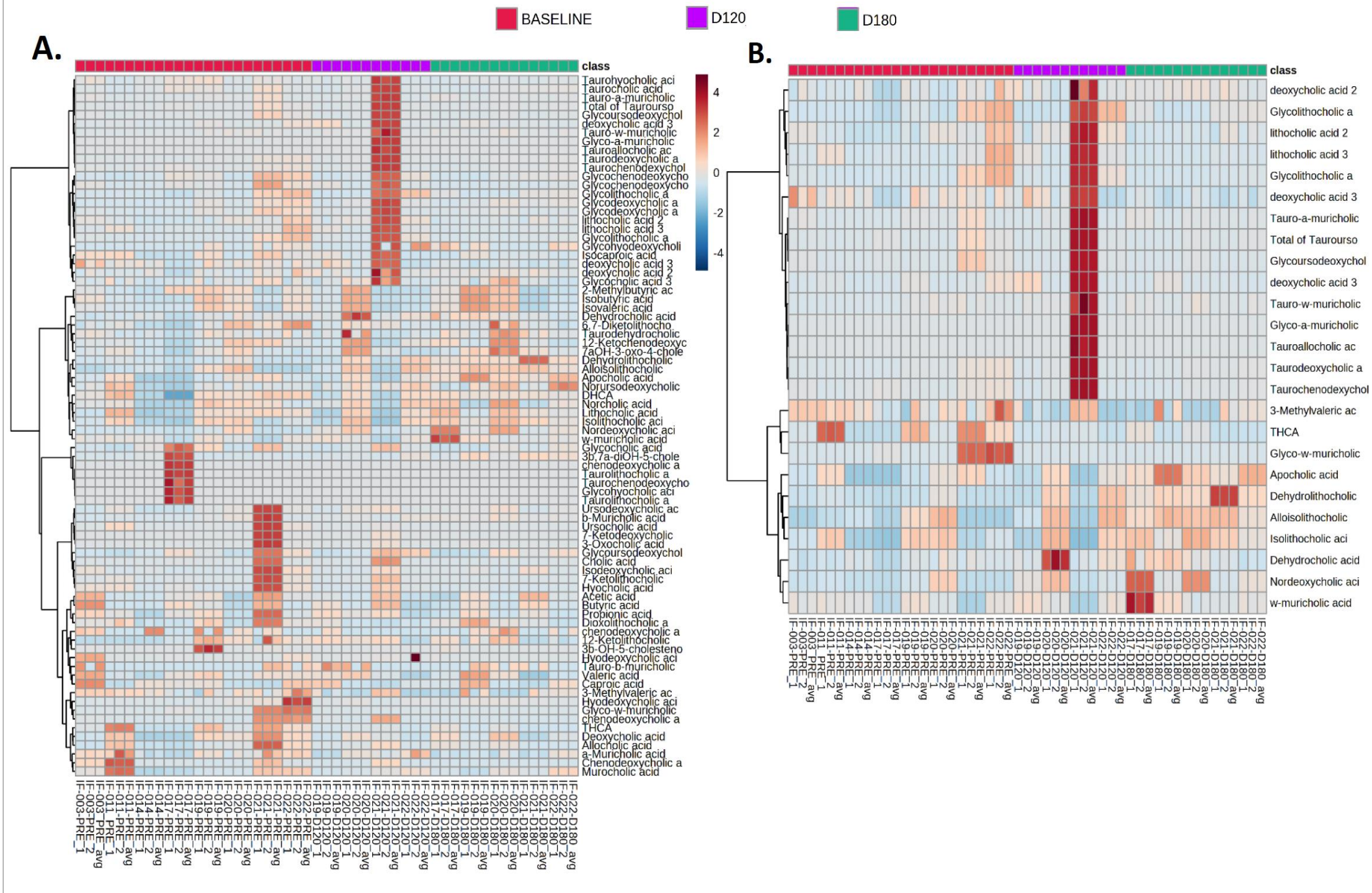
Autophagy

- induces autophagy via LC3B to suppress hepatocellular carcinoma in vivo and in vitro (Wang et al., 2017)

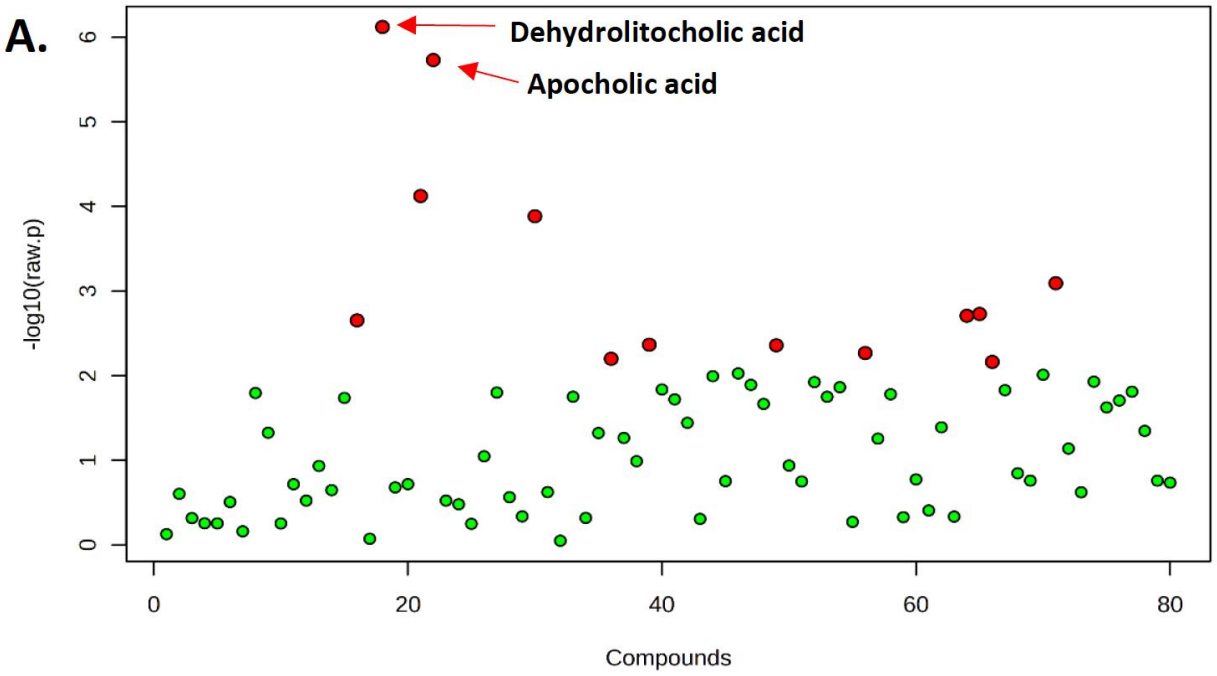


6-month intermittent fasting diet group

INCLUDED: 003, 011, 014, 017, 019, 020, 021, 022


BASELINE
D120
D180
A.
B.
class
Taurohyocholic aci
Taurocholic acid
Tauro-a-muricholic
Total of Tauroursо
Glycoursodeoxychol
deoxycholic acid 3
Tauro-w-muricholic
Glyco-a-muricholic
Tauroallocholic ac
Taurodeoxycholic a
Taurochenodexychol
Glycochenodeoxycho
Glycochenodeoxycho
Glycolithocholic a
Glycodeoxycholic a
Glycodeoxycholic a
lithocholic acid 2
lithocholic acid 3
Glycolithocholic a
Glycohyodeoxycholi
Isocaproic acid
deoxycholic acid 3
deoxycholic acid 2
Glycocholic acid 3
2-Methylbutyric ac
Isobutyric acid
Isovaleric acid
Dehydrocholic acid
6,7-Diketolithocho
Taurodehydrocholic
12-Ketochenodeoxyc
7aOH-3-oxo-4-chole
Dehydrolithocholic
Alloisolithocholic
Apocholic acid
Norursodeoxycholic
DHCA
Norcholic acid
Lithocholic acid
Isolithocholic aci
Nordeoxycholic aci
w-muricholic acid
Glycocholic acid
3b,7a-diOH-5-chole
chenodeoxycholic a
Taurolithocholic a
Taurochenodeoxycho
Glycohyocholic aci
Taurolithocholic a
Ursodeoxycholic ac
b-Muricholic acid
Ursocholic acid
7-Ketodeoxycholic
3-Oxocholic acid
Glycoursodeoxychol
Cholic acid
Isodeoxycholic aci
7-Ketolithocholic
Hyocholic acid
Acetic acid
Butyric acid
Propionic acid
Dioxolithocholic a
chenodeoxycholic a
12-Ketolithocholic
3b-OH-5-cholesteno
Hyodeoxycholic aci
Tauro-b-muricholic
Valeric acid
Caproic acid
3-Methylvaleric ac
Hyodeoxycholic aci
Glyco-w-muricholic
chenodeoxycholic a
THCA
Deoxycholic acid
Allocholic acid
a-Muricholic acid
Chenodeoxycholic a
Murocholic acid
4
2
0
-2
-4
deoxycholic acid 2
Glycolithocholic a
lithocholic acid 2
lithocholic acid 3
Glycolithocholic a
deoxycholic acid 3
Tauro-a-muricholic
Total of Tauroursо
Glycoursodeoxychol
deoxycholic acid 3
Tauro-w-muricholic
Glyco-a-muricholic
Tauroallocholic ac
Taurodeoxycholic a
Taurochenodexychol
3-Methylvaleric ac
THCA
Glyco-w-muricholic
Apocholic acid
Dehydrolithocholic
Alloisolithocholic
Isolithocholic aci
Dehydrocholic acid
Nordeoxycholic aci
w-muricholic acid

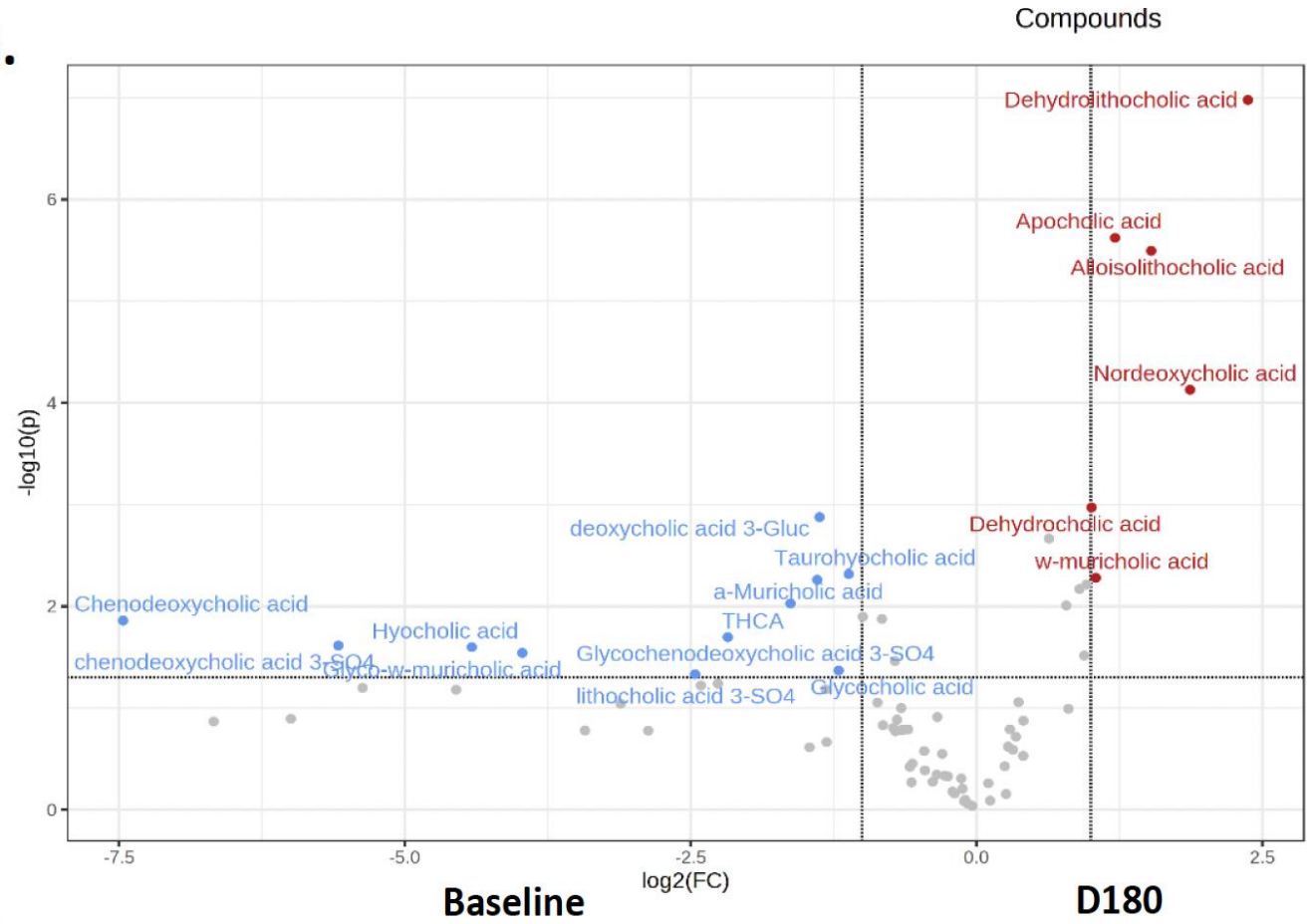
A.

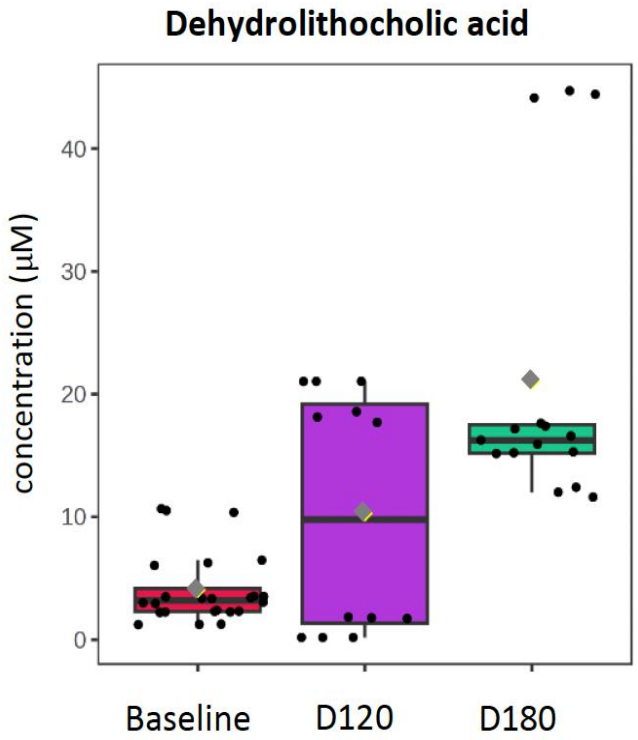


13 metabolites p<0.05*

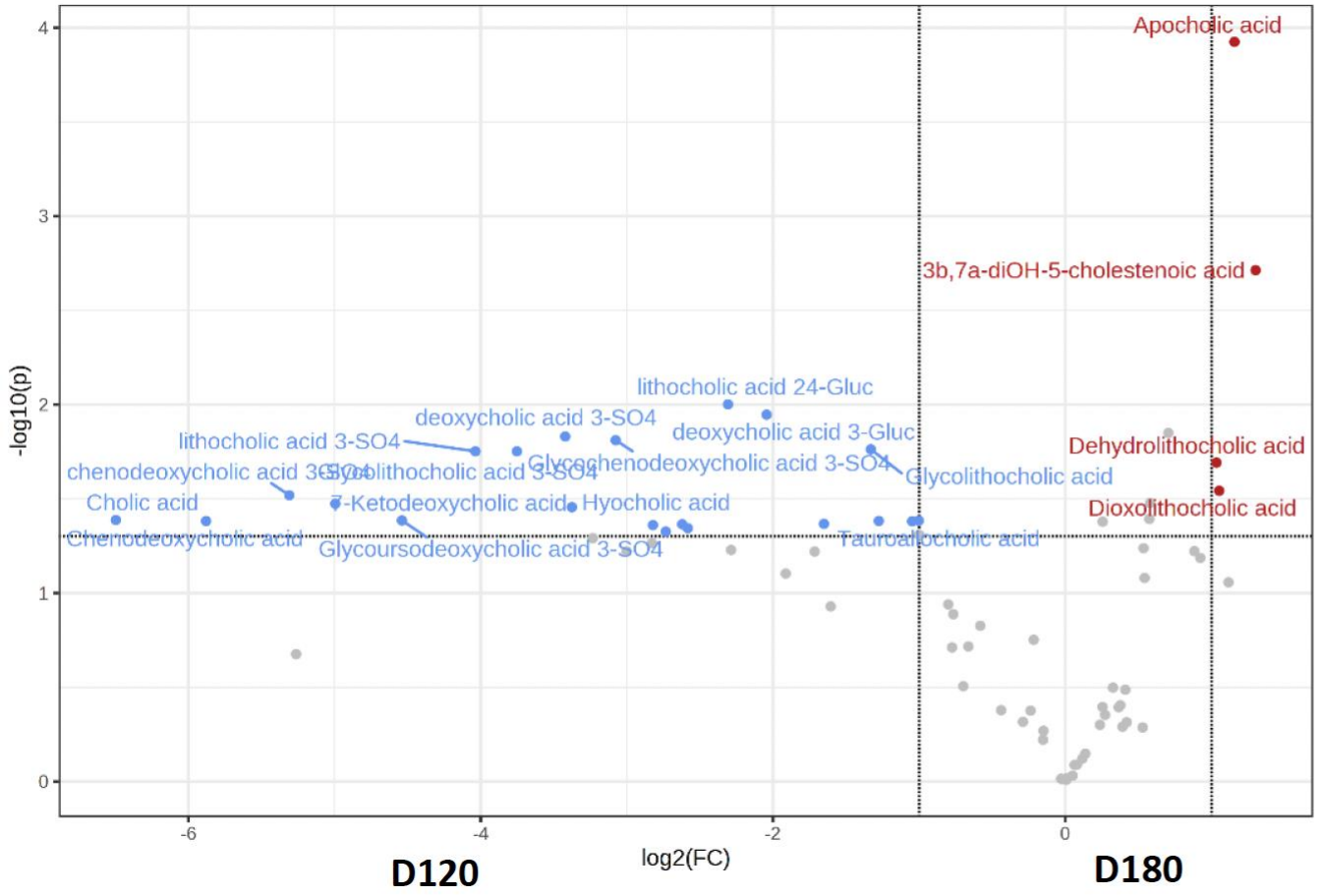
Metabolites	f.value	p.value	-log10(p)	FDR
Dehydrolithocholic acid	19.181	7.55E-07	6.1219	6.04E-05
Apocholic acid	17.591	1.86E-06	5.7308	7.43E-05
Alloisolithocholic acid	11.648	7.52E-05	4.1239	0.002005
Nordeoxycholic acid	10.839	0.0001304	3.8846	0.002609
deoxycholic acid 3-SO4	8.2901	0.000808	3.0926	0.012928
lithocholic acid 24-Gluc	7.1851	0.0018635	2.7297	0.022198
THCA	7.1204	0.0019588	2.708	0.022198
Dehydrocholic acid	6.9586	0.0022198	2.6537	0.022198
Glyco-a-muricholic acid	6.1176	0.0042993	2.3666	0.034991
Tauroallocholic acid	6.096	0.0043739	2.3591	0.034991
Tauro-w-muricholic acid	5.8342	0.005394	2.2681	0.039229
w-muricholic acid	5.6395	0.0063123	2.1998	0.042082
deoxycholic acid 3-Gluc	5.5347	0.0068721	2.1629	0.04229

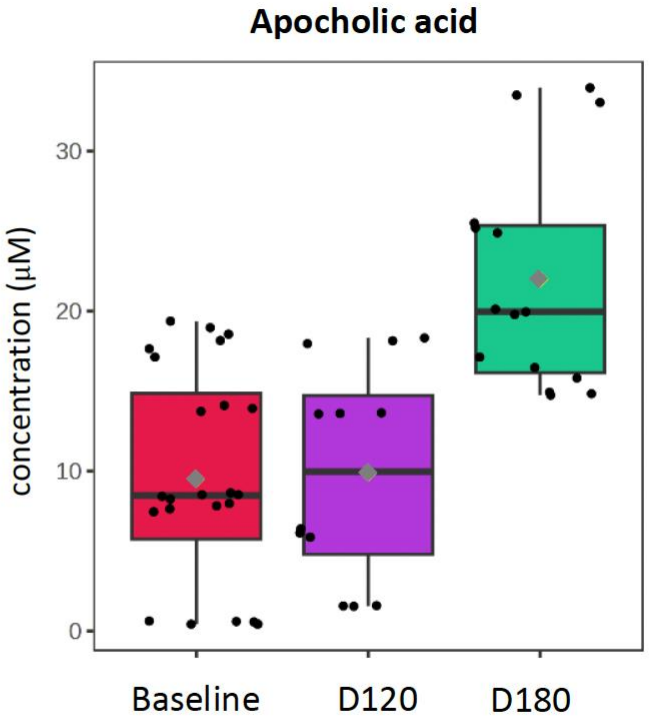
B.


Compounds
Dehydrolithocholic acid
Apocholic acid
Aboisolithocholic acid
Nordeoxycholic acid
Dehydrocholic acid
w-muricholic acid
deoxycholic acid 3-Gluc
Taurohyocholic acid
a-Muricholic acid
THCA
Glycochenodeoxycholic acid 3-SO4
Glycocholic acid
lithocholic acid 3-SO4
Chenodeoxycholic acid
Hyocholic acid
chenodeoxycholic acid 3-SO4
Glyco-w-muricholic acid
-log10(p)
log2(FC)
Baseline
D180
Status
DOWN
Non-SIG
UP


Dehydrolithocholic acid
concentration (µM)
Baseline
D120
D180

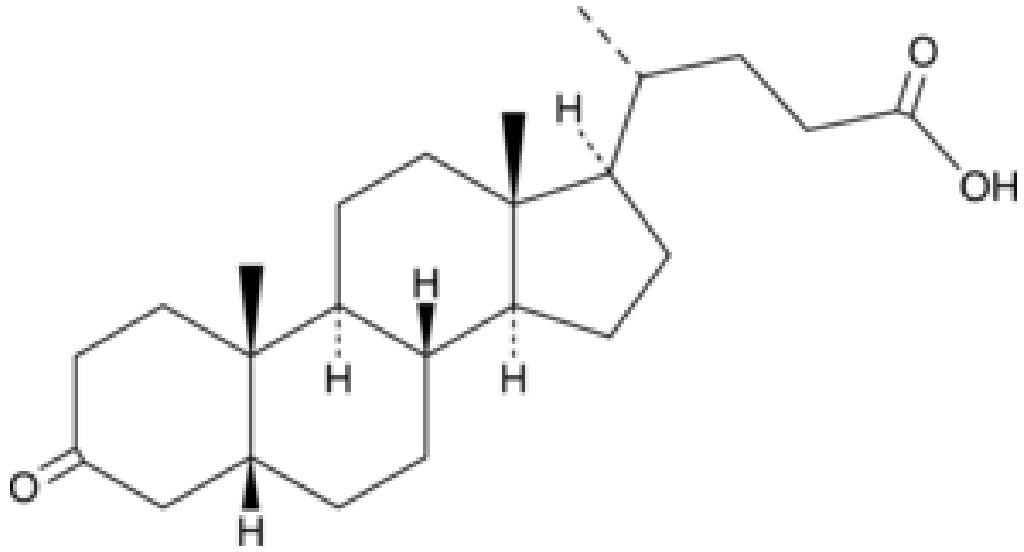
C.


Apocholic acid
3b,7a-diOH-5-cholestenoic acid
lithocholic acid 24-Gluc
deoxycholic acid 3-SO4
deoxycholic acid 3-Gluc
lithocholic acid 3-SO4
Glycochenodeoxycholic acid 3-SO4
chenodeoxycholic acid 3-Gluc
Glycolithocholic acid 3-SO4
Glycolithocholic acid
Dehydrolithocholic acid
Cholic acid
7-Ketodeoxycholic acid
Hyocholic acid
Dioxolithocholic acid
Chenodeoxycholic acid
Glycoursodeoxycholic acid 3-SO4
Taurolithocholic acid
-log10(p)
log2(FC)
D120
D180
Status
DOWN
Non-SIG
UP

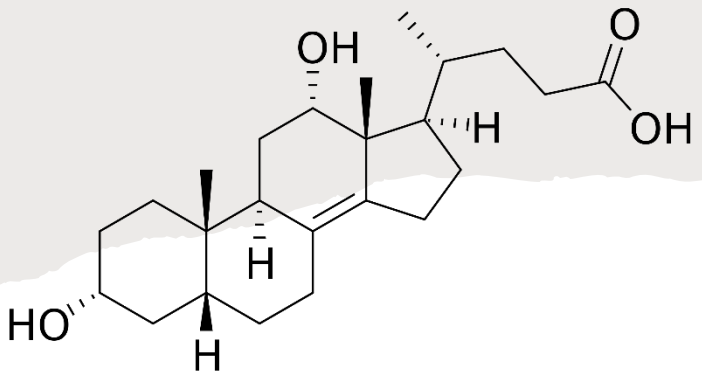

Apocholic acid
concentration (µM)
Baseline
D120
D180

Dehydrolithocholic acid

- major metabolite of lithocholic acid
- It is formed from LCA by the cytochrome P450 (CYP) isoform CYP3A4
- an agonist of G protein-coupled bile acid activated receptor 1 (GP-BAR1/TGR5) vitamin D receptor (VDR; EC50 = 3 µM), and farnesoid X receptor (FXR) in cell-based reporter assays (Mangelsdorf et al., 2002) Science & (Auwerx et al., 2008) J. Med. Chem.

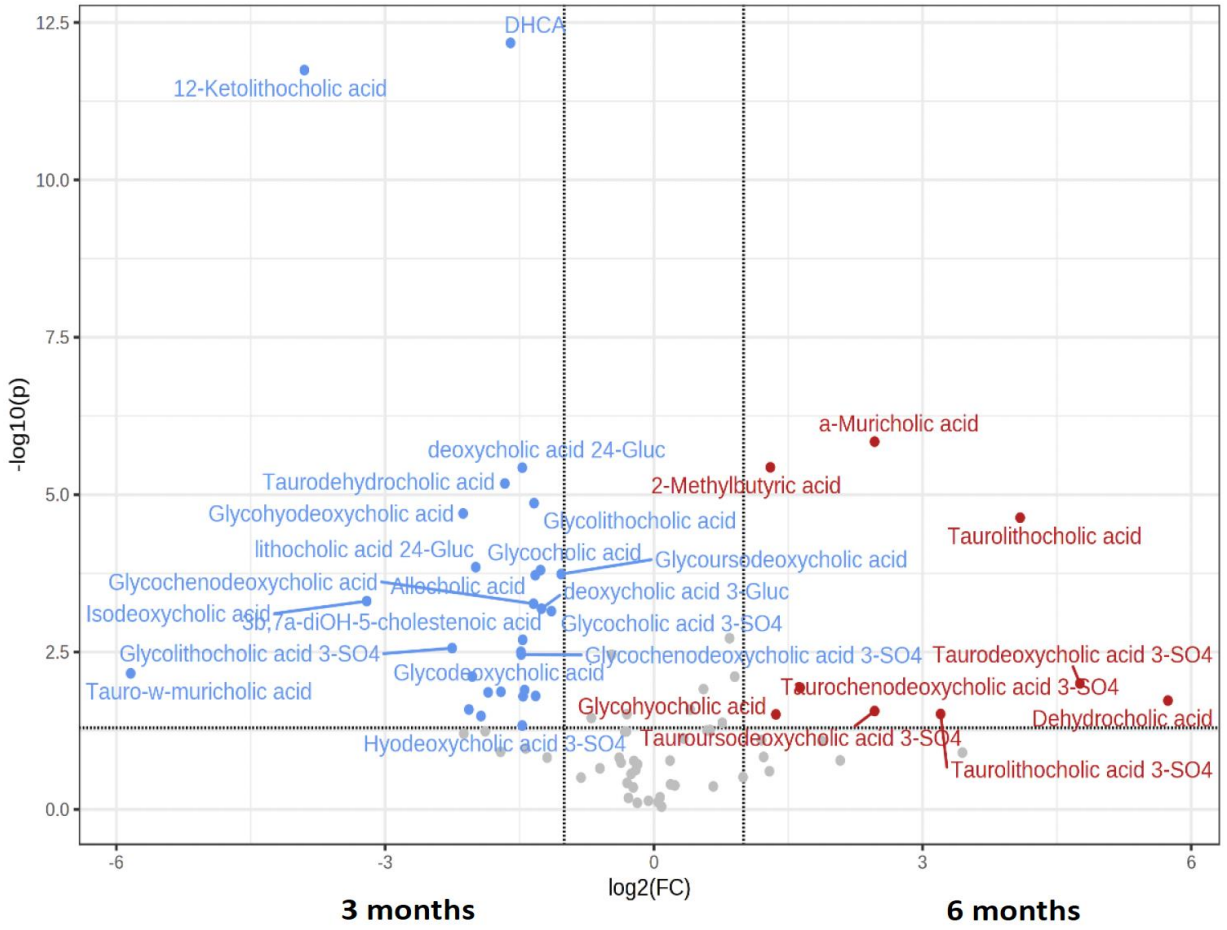


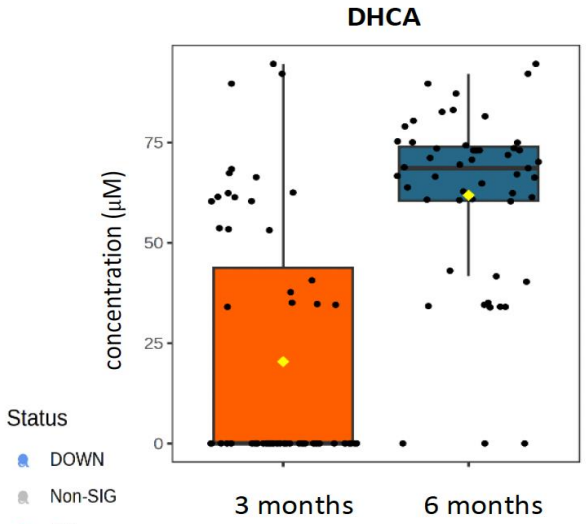
Apocholic acid

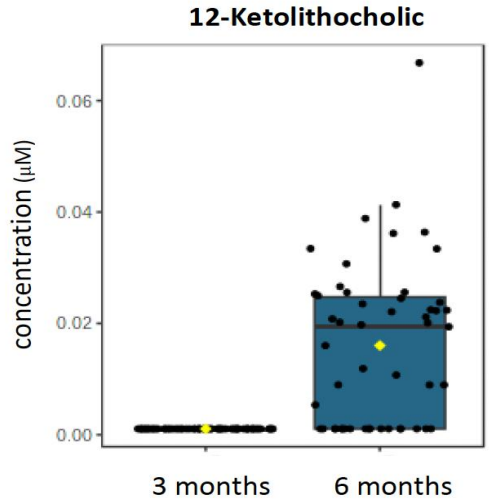


- Controversially been linked to carcinogenic activity, however limited reproducibility - nature
- Human studies
  - Metabolic profiling of BA in urine of patients with alcohol associated liver disease
    - **apocholic acid** had increased levels in patients with severe alcohol-associated hepatitis and might be **diagnostic of severe liver** disease – Hepatology Communications
  - Metabolic profiling of BA in serum with patients receiving various dosing of vitamin D
    - Metabolites associated with cholesterol metabolism and bile acid biosynthesis, such as 7-ketolithocholic acid, 12-ketolithocholic acid, **apocholic acid**, N-arachidene glycine and d-mannose 6-phosphate, were **significantly upregulated** in the group that had the **lowest vitamin D levels** (<30ng/mL) – Open Medicine
- Mouse study
  - diarrhea induced by capecitabine (sample: serum)
  - For mice that ultimately developed diarrhea, **apocholic acid**, isodeoxycholic acid, and 7-ketodeoxycholic acid exhibited the **largest change in concentrations** compared with their baseline concentrations - Journal of Pharmaceutical and Biomedical Analysis

3 month vs. 6 month

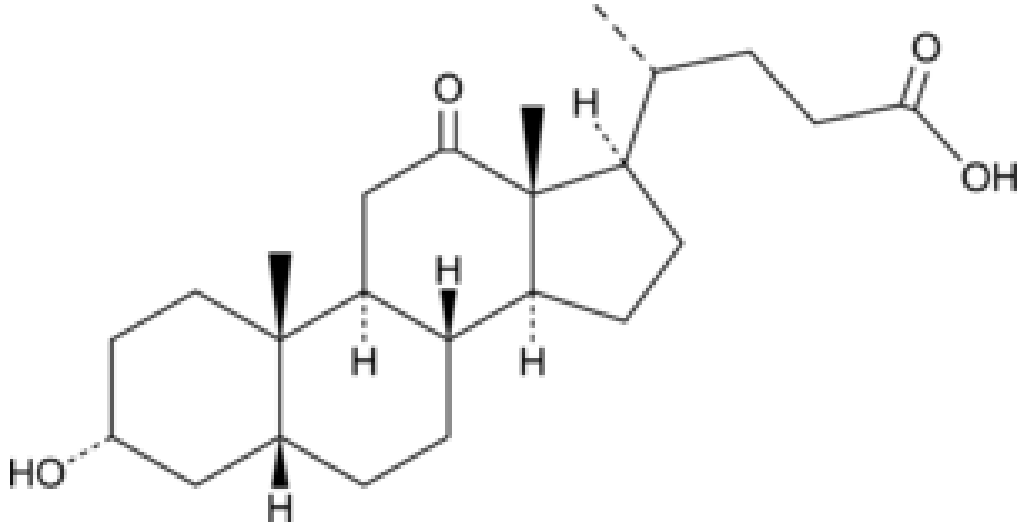






12-keto litocholic acid

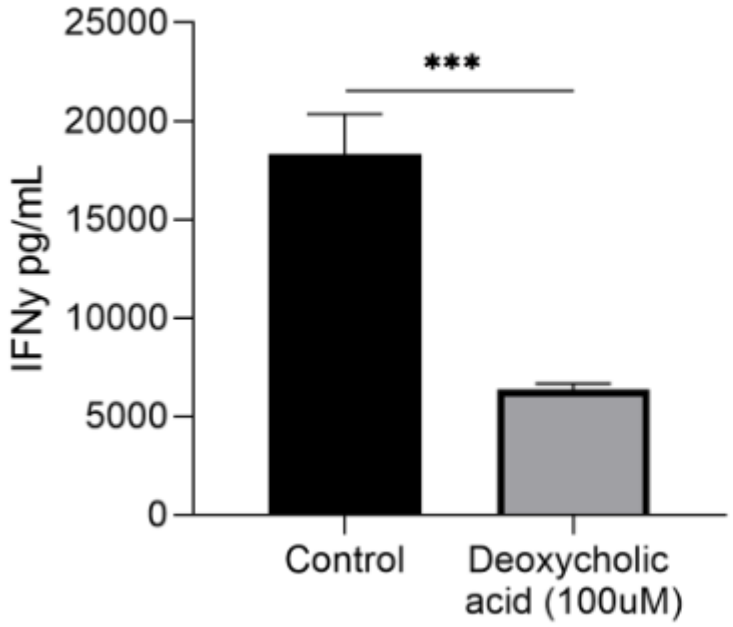
- metabolite of the secondary bile acid deoxycholic acid
- Fecal levels of 12-keto lithocholic acid are elevated in rats fed a diet supplemented with cholic acid or DCA compared with rats fed a control diet or a diet supplemented with cholesterol, hyodeoxycholic acid or lithocholic acid – (Mutai et al., 1979) Microbiol. Immunol.
- 12-keto Lithocholic acid levels are increased in duodenal bile from patients with type 2 diabetes.- (Sjovall et al., 1988) Eur J Clin Invest.

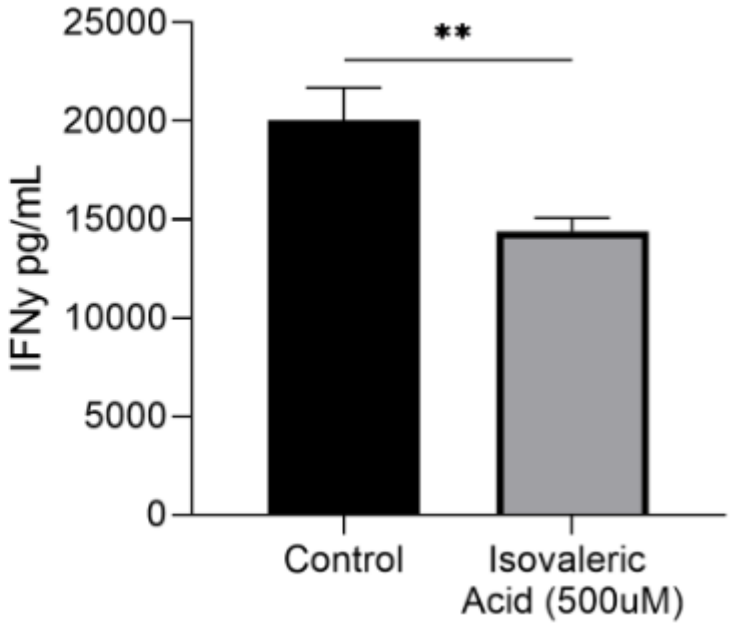


SCFA on Tregs and CD8+ T cells

- Unconfirmed role of SCFA's positive effects on Tregs
  - Although there is conflicting literature on the role of SCFAs in mediating the effects of the microbiota on Treg cells. SCFAs are taken up by colonic epithelial cells and diffuse through the epithelium into the colonic lamina propria, where they mediate their effects directly on DCs and T cells - Nature (Honda et al., 2016)
  - Show no effects of SCFAs on Tregs – Science Advances (Benoist et al., 2015)
- In vitro treatment of pentanoate and butyrate with CD8+ T cells and CAR T cells increases their function by elevating IFN-γ and CD25 - Nature (Visekruna, 2021)

Some metabolites are shown to be immunosuppressive





experiment performed by Sarah MacPherson

What is happening?

- 3-month cohort: UDCA plummets ($p<0.001$) at D120, while autophagy levels drop (D150). Glycolithocholic acid steadily drop through fast and post-fast.
- 6-month cohort: UDCA does not go significantly up/down.. autophagy decreases but continues to plateau
- Both cohorts have a unique set of metabolites that are significant, with not much overlap

What's next?

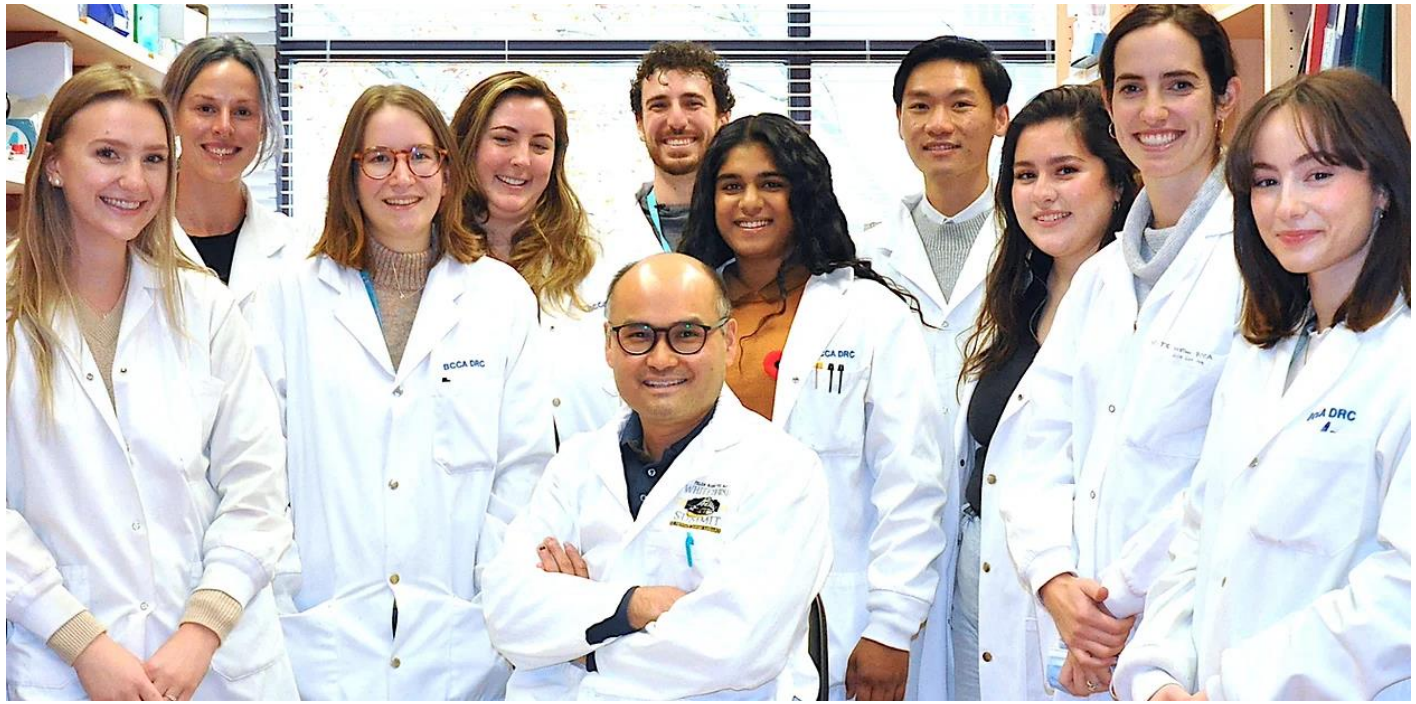
- Functional in vitro experiments of SCFAs/BAs on T cells

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BC CANCER FOUNDATION

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